(CSE403) Random Day Quiz 1

Student Name:

Student ID:

1. The output of an MLP depends on:
    a. Weights b. Biases c. Activation functions d. All of the above

2. Which statement best describes backpropagation?
    a. Updates weights from input to output directly
    b. Computes gradients layer by layer from output to input
    c. Randomly changes weights until loss decreases
    d. Propagates inputs forward through the network

3. In a convolutional layer, the kernel (or filter):
    a. Scans over the input to extract local patterns
    b. Stores class labels
    c. Applies dropout regularization
    d. Performs global matrix multiplication

4. What is the main role of pooling layers in CNNs?
    a. Increase resolution
    b. Reduce spatial size and computation
    c. Add more channels
    d. Normalize the input

5. Which best explains weight sharing in CNNs?
    a. Each neuron has unique weights
    b. The same filter weights are applied across spatial locations
    c. Filters are reinitialized per image
    d. Filters only work for one region

6. Which type of data are RNNs specifically designed to handle?
    a. Randomized data
    b. Image data
    c. Sequential or time-dependent data
    d. Static tabular data

7. In an RNN, the hidden state at time t depends on:
    a. Only the input at time t
    b. The input and the previous hidden state
    c. All future inputs
    d. The output layer

8. Which of the following problems is common in vanilla RNNs?
    a. Vanishing and exploding gradients
    b. Weight sharing
    c. Dropout overfitting
    d. Activation saturation

9. LSTM networks are designed to:
    a. Ignore past information
    b. Solve vanishing gradient problems using memory gates
    c. Increase training speed
    d. Perform convolution on sequences

10. In an MLP, each neuron in one layer is connected to all neurons in the next layer.

True False

11. Dropout randomly deactivates some neurons during training to prevent overfitting.

True False

12. ReLU activation can cause the vanishing gradient problem more severely than sigmoid.

True False

13. In CNNs, using a larger stride increases the output feature map size.

True False

14. In an RNN, parameters (weights) are shared across all time steps.

True False

15. The vanishing gradient problem means the gradients become too large during training.

True False

(CSE403) Random Day Quiz 2

Student Name:

Student ID:

1. Which statement best explains why Transformers enable fast training?
   a. Transformers remove linear layers
   b. Transformers eliminate backpropagation
   c. Transformers enable parallel computation by removing recurrence
   d. Transformers do not use embeddings

2. In a Transformer, positional encoding is used to:
   a. Replace self-attention
   b. Add order information to the input embeddings
   c. Increase vocabulary size
   d. Control dropout rate

3. In a Transformer, self-attention computes relationships between:
   a. Only adjacent words
   b. All words in the sequence
   c. Only encoder outputs
   d. Token positions only

4. Which statement best describes the role of scaling in Transformer attention?
   a. Prevents the model from attending to too many positions
   b. Avoids extremely large attention scores that make softmax unstable
   c. Reduces the number of attention heads
   d. Makes positional encoding unnecessary

5. Multi-head attention in Transformers allows the model to:
   a. Increase sequence length
   b. Look at different types of relationships in parallel
   c. Replace embeddings
   d. Freeze gradients

6. A Transformer decoder uses masked self-attention to:
   a. Speed up training
   b. Prevent the model from looking at future tokens
   c. Reduce model parameters
   d. Increase embedding dimension

7. Why does cross-attention use keys and values from the encoder instead of the decoder?
   a. Because encoder representations contain information about the full input sequence
   b. Because decoder states do not have embeddings
   c. Because keys cannot be computed at the decoder
   d. Because positional encoding must only be applied once

8. In a Graph Neural Network (GNN), a node updates its representation by:
   a. Aggregating information from random nodes
   b. Aggregating information from its neighbors
   c. Ignoring neighbors
   d. Using only the adjacency matrix

9. In a GNN, deeper layers allow a node to gather information from:
   a. Only itself
   b. Only direct neighbors
   c. Nodes that are k hops away
   d. The whole graph immediately

10. In GNN neighborhood aggregation, dividing by |N(v)| (the number of neighbors) ensures that:
    a. Node degrees become irrelevant
    b. All neighbors contribute equally regardless of degree
    c. The model trains faster
    d. Only the target node's own features matter

11. A Graph Neural Network can generate embeddings for unseen nodes because:
    a. It memorizes all nodes during training
    b. Aggregation functions depend only on local neighborhoods
    c. It uses recurrence
    d. It requires node IDs to be fixed

12. One of the key advantages of Transformers over RNNs is the shorter path length between tokens.

    True False

13. The feed-forward network in a Transformer applies the same transformation independently to every position.

    True False

14. A standard MLP is naturally permutation-equivariant for graph inputs.

    True False

15. GNNs repeatedly apply neighborhood aggregation layers to compute node embeddings.

    True False